

Campus Collaboration Finder: A Skill-Based Project Matching Platform for University Students

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Abstract—The Campus Collaboration Finder is a web-based platform that connects university students for collaborative projects through skill-based matching. Students can create profiles, post projects with required skills, apply to projects, and receive AI-powered recommendations. The system is built using Flask (Python) backend, MySQL database, and a responsive frontend. It provides full CRUD functionality along with a smart matching algorithm that calculates skill match percentage.

Index Terms—Skill-based Matching, Project Collaboration, Recommendation System, Flask API, MySQL, Web Application

I. INTRODUCTION

Finding suitable team members for academic projects is often challenging for university students. This project aims to solve this issue by developing a centralized platform that matches students with projects based on their skills.

II. SYSTEM DESIGN

A. Entity Relationship Diagram

Figure 1 shows the Entity Relationship Diagram of the system.

B. Database Tables

The system uses six main tables: Students, Skills, Student_Skills, Projects, Project_Skills, and Applications.

III. IMPLEMENTATION AND USER INTERFACE

A. Students Management

Figure 2 shows the Students management interface.

B. Projects Management

Figure 3 shows the Projects section.

C. Applications Management

Figure 4 shows the Applications panel.

D. AI Recommendation System

The system includes an intelligent recommendation feature. Figure 5 shows the AI Match page.

IV. CONCLUSION

The Campus Collaboration Finder successfully provides a complete solution for campus project collaboration with skill-based matching.

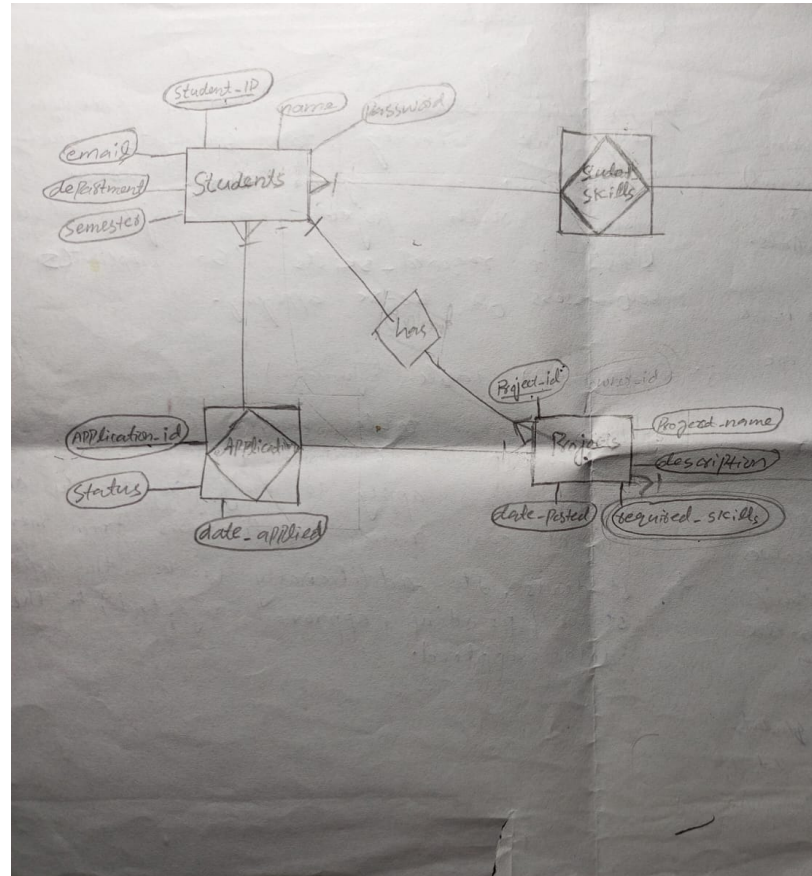


Fig. 1. Entity Relationship Diagram (ERD) of Campus Collaboration Finder

REFERENCES

- [1] Flask Documentation, <https://flask.palletsprojects.com/>
- [2] MySQL Official Documentation, <https://dev.mysql.com/doc/>

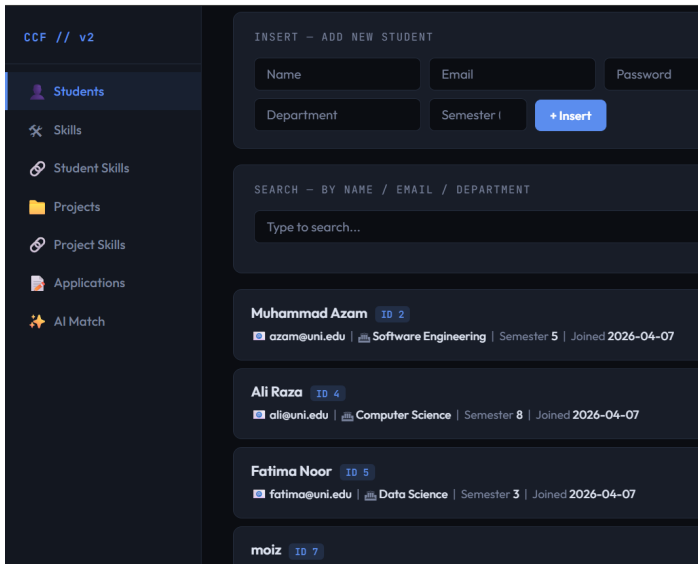


Fig. 2. Students Management Interface

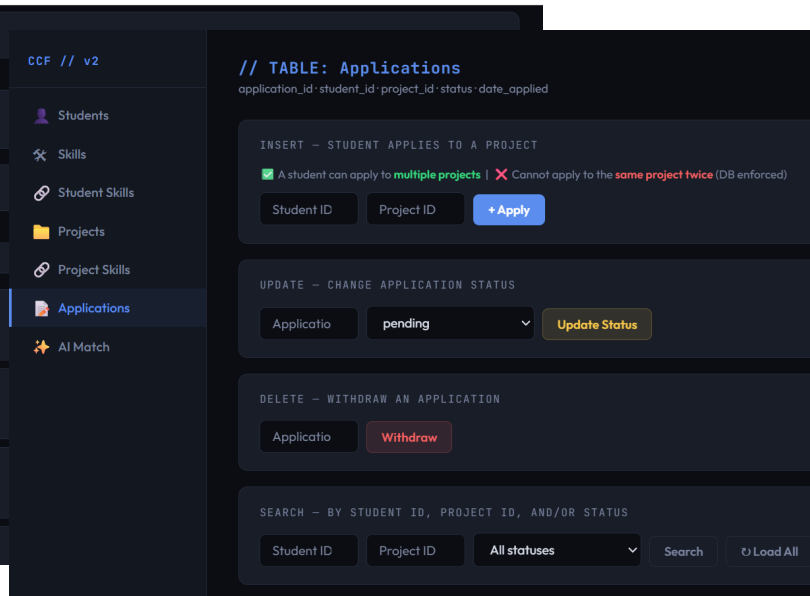


Fig. 4. Applications Management Interface

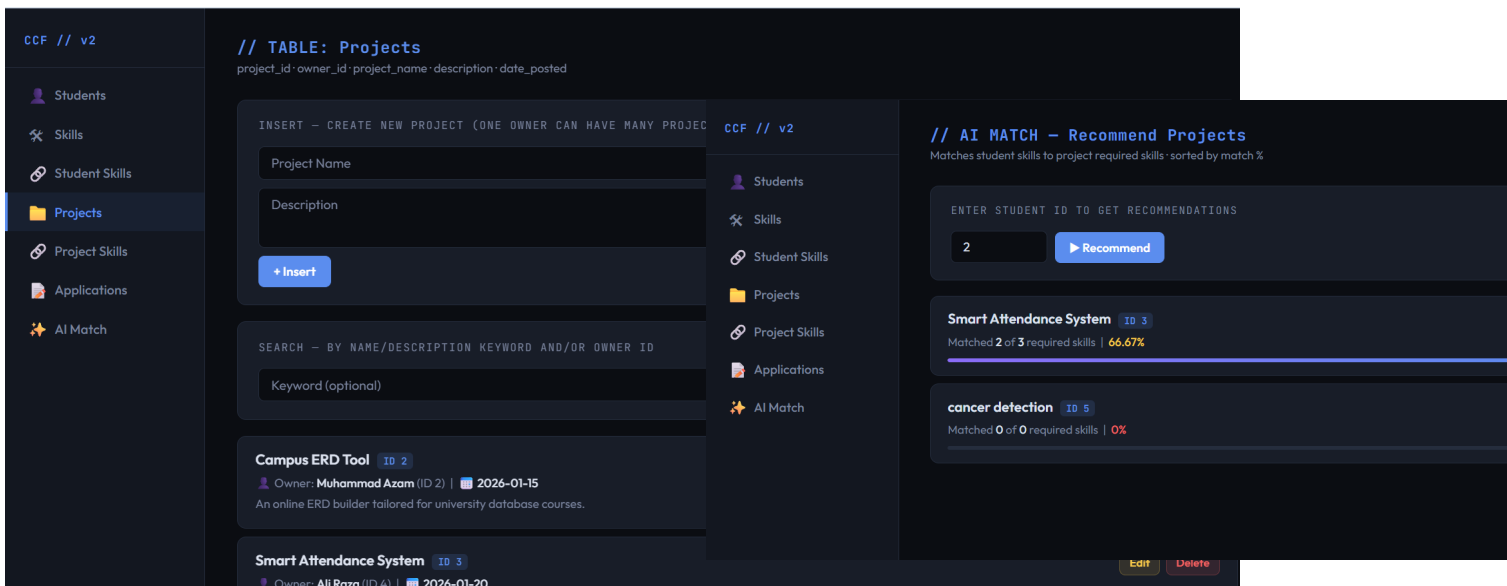


Fig. 3. Projects Management Interface

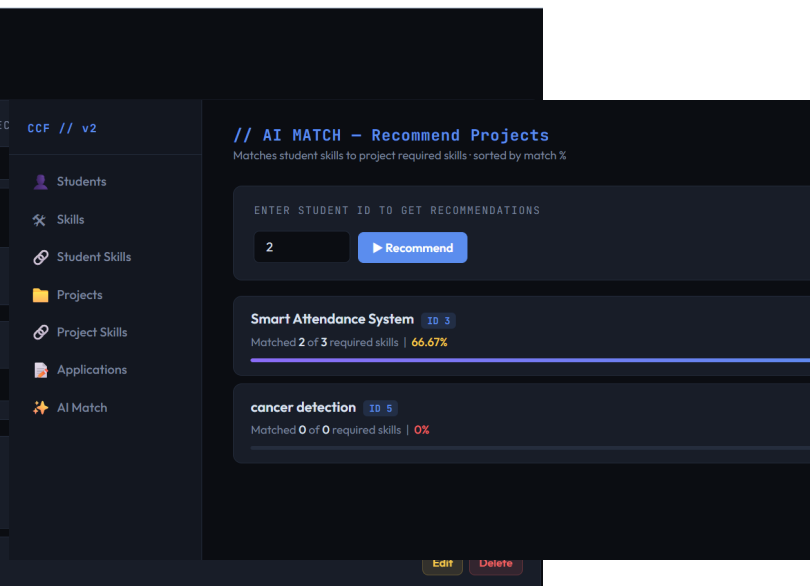


Fig. 5. AI-Powered Project Recommendation Interface